

APP DEV PROJECT REPORT

1. Student Details

Name : Muskan Jadon

Roll No : 24f2005143

Email : 24f2005143@ds.study.iitm.ac.in

About me : I am a Student of IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title : Trekking Management Application

Problem Statement:

Develop a centralized web-based system to efficiently manage trekking activities, bookings, staff coordination, and user participation by replacing manual management processes.

Approach:

Develop a Flask-based role-driven web application that digitizes trekking management. The system will provide separate dashboards for Admin, Trek Staff, and Trekkers to manage treks, staff assignments, bookings, and trekking history. SQLite will be used for data storage with role-based authentication, booking management, and overbooking prevention.

3. AI/LLM Declaration

AI tools such as ChatGPT were used for assistance in code structuring, debugging suggestions, and documentation formatting. The extent of AI/LLM usage is around 15-20%, limited to code suggestions and documentation formatting. The core logic, implementation, and integration of the application were developed manually.

4. Technologies and Frameworks Used

Technology / Library	purpose
Flask	core backend framework
SQLAlchemy	object Relational Mapper for SQLite Database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap5	Fronted styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing data
Werkzeug	Password hashing

5. Database Schema / ER-Digram

Tables:

1. **User** -Stores login and role information for Admin, Trek Staff, and Trekkers. (id, email, password, role,is_blacklisted)
2. **Staff profile** -Stores additional details of trek staff. (id, user_id, fullname, contact, approval_status,is_blacklisted)
3. **Trek** -Stores information about trekking events. (id, trek_name, location, difficulty, duration, available_slots, assigned_staff_id, status, start_date, end_date)
4. **Booking** -Stores trek booking records.
(id, user_id, trek_id, booking_date, status)

Relationships:

One-to-One → User → StaffProfile

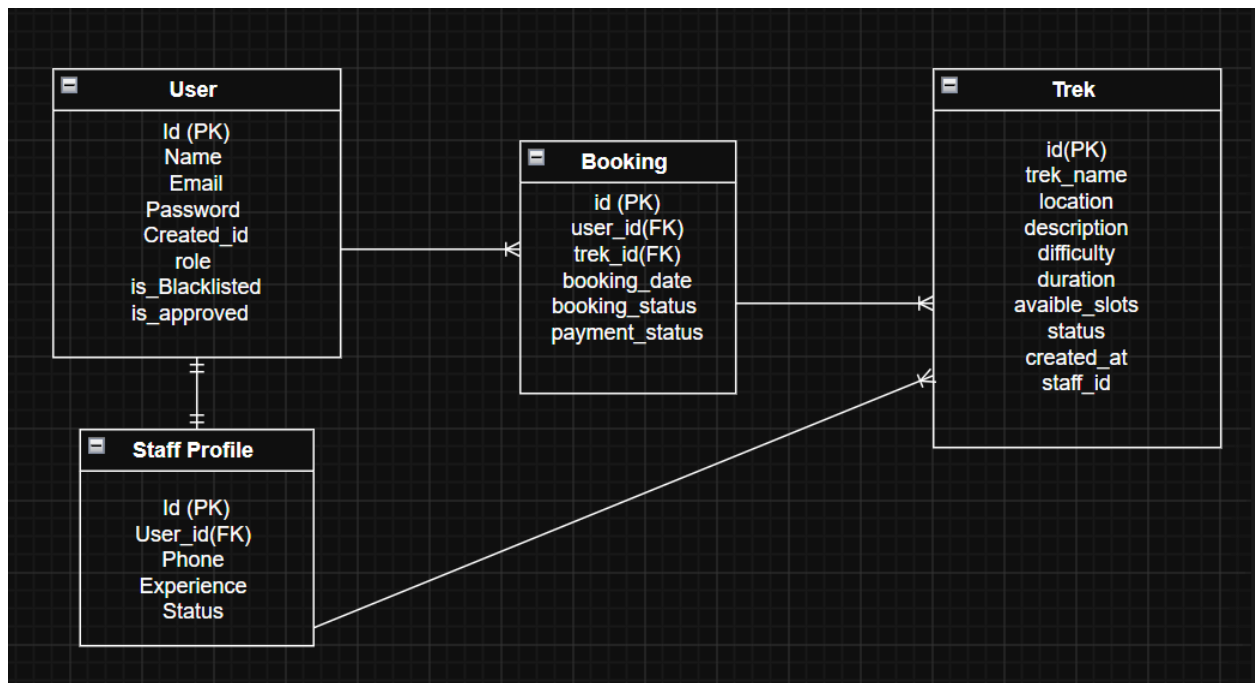
One-to-Many → User (Trekker) → Booking

One-to-Many → Trek → Booking

One-to-Many → StaffProfile → Trek (One staff can manage multiple treks)

Many-to-One → Booking → User

Many-to-One → Booking → Trek



6. Architecture and Features

Architecture Overview

- [app.py](#) - main flask application entry point
- /models - SQLAlchemy database models
- /routes - Flask blueprints for admin and student routes
- /templates - Jinja2 HTML templates for UI

Implemented Features

- Admin login and dashboard
- Trek Staff and User (Trekker) registration and login system
- Admin can create, update, and manage treks

- Admin can approve and assign Trek Staff to treks
- Users can view, search, and book available treks
- Trek Staff can manage assigned treks and update trek status
- Booking history and participant management
- Responsive UI using Bootstrap
- Secure authentication using Flask-Login
- Overbooking prevention through slot availability checks

Video Drive

Link: <https://drive.google.com/file/d/1Jub- G ONJ hz LCvLY I KRXL1nDv Uz/view?usp=drive link>